Balanced collision packing for exact counting

A new quantitative refinement of the October 2026 ETH algorithm

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Abstract

We review the seven ETH-related proposal archives and the separate 49-page manuscript present in a pinned snapshot of *ProveIt*, and improve the strongest comparable explicit contraction bound. The new ingredient is a deterministic two-stage labeling of a bipartite multigraph: balance row degrees first, then choose column labels by exact rational conditional expectation. This permits nearly equal loads in the pair-label classes under $M \gg D_{\mathrm{arr}}\Delta \operatorname{polylog} s$, instead of the earlier sufficient condition $M \gg D_{\mathrm{arr}}^2\Delta$. It also removes the leading factor three in packet padding. An additive logarithmic precision guard replaces the source's full extra taper block. Finally, an integer capacity rule uses the available packing budget without leaving fixed asymptotic slack.

Combined with the already proposed exact Hermite taper and factorial Cauchy modes, these changes give the normalized tensor-contraction bound

$$(s+\mathcal{L})^{O(1)}2^{(C_*+o(1))s/\sqrt{\log_2 s}}, \qquad C_* = \frac{9\pi}{4\ln 2}\sqrt{15\log_2 54} = 94.7482022595\ldots,$$

in polynomial space. The strongest numerically instantiated prior coefficient is $648.4410666487\ldots$; its combination with the earlier Cauchy modes gives $374.3776243833\ldots$. The prior tunable schedule family approaches approximately 638.52749, or 368.65402 after those Cauchy modes; the new result improves on this stronger comparison as well. These are comparisons of proved upper-bound certificates, not runtime measurements or lower bounds for previous algorithms. The general fixed-width CNF order remains $\operatorname{poly}(n,m)2^{O(n\log\log n/\sqrt{\log n})}$. The independent 49-page manuscript has no explicit comparable coefficient.

The report proves the new statements, identifies all reused ingredients, and supplies exact-arithmetic packing and schedule prototypes with finite checks. It does not supply a complete SAT solver or a Lean formalization.

Source pins. ProveIt: `aa6933442eb936962de79ba6d321cd0391c450bd`.
OpenAI math: `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`.
Scope: the three ETH intake commits dated October 26 in Pacific time, all seven proposal archives they introduce, and the two accompanying PDFs.

Placement and summary follow-up checked through `45233238aebf`.

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1 Result and comparison with the reviewed work

The requested improvement is possible at the level of an explicit leading exponential coefficient. This report does not establish a smaller logarithmic order for unrestricted 3SAT. That distinction is central: C_* is a coefficient for a specified normalized tensor-network problem, and is not automatically a coefficient multiplying the number of SAT variables.

The October 25 OpenAI manuscript [1] states, for every fixed clause width k,

$$T_{\text{general}}(n,m) = (n+m)^{O_k(1)} 2^{O_k(n \log\log n/\sqrt{\log n})}, \tag{1.1}$$

$$T_{\text{fixed}}(n,m) = (n+m)^{O_{k,B}(1)} 2^{O_{k,B}(n/\sqrt{\log n})}, \qquad m \le Bn, \quad B \text{ fixed}. \tag{1.2}$$

Its scope document [2] distinguishes these manuscript statements from the released Lean endpoint, whose general exponent contains $(\log\log n)^2$. This report compares against the sharper manuscript construction. Removing a factor already removed in its appendix would not be a new result.

1.1 Exactly what was reviewed

At the ProveIt pin above, the relevant intake commits are `a505bd768bce`, `64b3202ee84f`, and `8afdbbea48d6`. Their complete identifiers, changed-file lists, archive sizes, and hashes are recorded in the accompanying provenance files. The first commit brings three proposal archives and two PDFs; the second brings three more archives; the third brings the seventh archive. The later placement and summary through `45233238aebf` were also checked: the seven reports and their auxiliary LaTeX files match the reviewed archive bytes. Their current project is `Computability/SubexponentialSAT`.

Packet / manuscript	Strongest relevant established contribution	Limit of the claim
Contraction improvements [3]	Exact taper, leading precision $6MD/u$, revised schedule with coefficient $648.4411\ldots$, a tunable family approaching $638.5275\ldots$, and streamed packet coefficient extraction	Same logarithmic order
Factorial Cauchy modes [4]	Mode weights $1 + b + 2v_2(b!)$, giving a $1/\sqrt{3}$ leading entropy factor	An exponential coefficient improvement
Precision refinement [5]	Exact recovery loss, optimal taper within a specified functional, and a structured-mode representation	Its $n \log\log n/\log n$ target requires an unproved summation hypothesis
Scalar parity refinement [6]	Scalar parity lift and a smaller relevant degree cap	A second-contraction acceleration is conditional
Density improvement [7]	Density-aware sparse checks and exponent $n \min\{\log(2+m/n), \log\log n\}/\sqrt{\log n}$	Improvement depends on clause density
Density refinement [8]	A closely related density-sensitive bound and precision refinements	Same unrestricted logarithmic order
Asymptotic analysis [9]	Audit, taper/Newton refinements, and precise conditional research targets	Explicitly does not establish a stronger unconditional worst-case order
Separate `eth.pdf` [10]	A distinct eight-square/clique-network approach, with an absolute hidden constant and the same fixed-k general logarithmic order	No explicit final coefficient is available for numerical dominance

The uploaded `paper.pdf` is the original OpenAI paper, rather than an eighth refinement archive. The separately uploaded `eth.pdf` is titled *A $2^{o(n)}$ Time Algorithm for 3-SAT*, attributed in its title page to

Claude / Anthropic and dated October 26, 2026. Its theorem states

$$2^{O\left(\min\left\{\frac{k^2 n \log\log n}{\sqrt{\log n}}, \frac{km}{\sqrt{\log m}}\right\}\right)}.$$

This establishes the scope of the comparison, not independent verification of that manuscript's theorem. Its appendix asserts a formalization, but the public-proof link is a placeholder in the inspected PDF; no external formalization of that manuscript was checked here.

1.2 The new and reused ingredients

There are three new algorithmic changes:

1. A two-stage, deterministically derandomized labeling that improves the power of the pair-class count in the sufficient packing condition.
2. An additive guard of $O(\log\log s)$ bits per recursion level, which preserves precision without duplicating the entire taper budget.
3. An integer capacity rule which pays the guard explicitly and attains the limiting leading coefficient with a single algorithm.

The nearly lossless packet count follows from the new labeling and exact padding arithmetic. The exact Hermite taper, factorial Cauchy basis, deformation identities, and recursive tensor factorization are reused and attributed; they are not claimed as new here.

Comparable normalized upper-bound certificate	Base-two coefficient
Prior numerically instantiated schedule	$648.4410666487\ldots$
That schedule combined with prior factorial Cauchy modes	$374.3776243833\ldots$
Prior optimized-family boundary, with Cauchy modes	≈ 368.65401812
New packing and guard, simple fixed $r = 64$ schedule	$95.4255527758\ldots$
New packing and guard, fixed $r = 54$ schedule	$94.7537015001\ldots$
New adaptive integer-cap schedule	$\mathbf{94.7482022595}\ldots$

The ratios $6.8438\ldots$, $3.9513\ldots$, and $3.8909\ldots$ compare coefficients in upper bounds, using the three relevant preceding baselines. They do not mean that a solver runs that many times faster. Nor do two upper bounds prove a lower bound on the old implementation. The alternate manuscript's unknown constant cannot be ordered using this table.

For a comparison with the entire prior tunable family, its factorial-adjusted boundary coefficient is

$$C_{\mathrm{old},3}(r) = \frac{2\pi\sqrt{160\log_2 r}}{\ln 2\left(1-(6/r)^{2/3}\right)^{3/2}}, \qquad r > 6.$$

This formula follows from the schedule family already described in [3], after applying [4]. The best boundary value is approximately 368.65402 near $r = 251$; strict feasible parameters approach the boundary from above. Numerical optimization is not needed to prove that the new certificate is smaller throughout that family: $\log_2 r > 2$, $\pi > 3$, and $\ln 2 < 1$ give

$$C_{\mathrm{old},3}(r) > 6\sqrt{320} > 107 > C_*.$$

This comparison concerns the stated upper-bound family, not a lower bound on the actual running time of any implementation.

2 Normalized interface and inherited identities

Let $s \geq 2$ be an integer global size bound and $\mathcal{L}$ the input description length. The input is a bipartite Boolean tensor network with at most s edges, maximum degree at most

$$\Delta = \left\lceil s^{9/10} \right\rceil,$$

and at most polynomially many described vertices. Its entries lie in

$$\mathcal{B}_{P_0,\ell} = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell}), \qquad 1 \leq P_0, \ell \leq s.$$

For fixed constants, an entry is computable in $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ time and polynomial space. These are the normalized hypotheses used by the prior explicit certificate. A source theorem allowing Cs initial edges requires the corresponding change of normalization; the number C_* below assumes at most s.

The original construction [1] works over an unramified coefficient extension $\mathcal{O}$ with residue field of size $2^{4\lambda}$, where

$$\lambda = \lceil \log_2 s \rceil.$$

Arithmetic, coordinate precision, and truncated V-coordinates are retained. Tensor descriptions also retain their *creation precision*: each entry is canonically lifted from its original creation-time residue. Lifting is not assumed to commute with multiplication or addition.

At an active recursive call there are M real edges. Packets have $L = 2^a$ slots. For the grid

$$S = \{-L, \ldots, L-1\}, \qquad \mathcal{C}(T) = \prod_{q \in S}(T-q), \qquad \alpha = 2^{a+1},$$

the original kernel is

$$W_{xy}(T) = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)(T-(y-x))}.$$

The denominator cancels a numerator factor; $W(0) = I$ on real ports. The source deformation $F(T)$ therefore has $F(0)$ equal to the desired contraction and degree at most

$$\deg F \leq M(2L-1). \tag{2.1}$$

The source proves coefficientwise integrality of $F(t+\alpha z)$, the Hermite recovery formula, and the source/target factorization after crossing modes are fixed.

The proofs below modify the packing and parameter interfaces to those identities. They do not presuppose a new fast algorithm for summing a general compressed tensor representation.

3 A new deterministic near-balanced packing lemma

The graph in this section is a bipartite multigraph. Parallel edges are counted with their multiplicity. Write

$$Q = r^a, \qquad D_{\mathrm{arr}} = Q^2, \qquad \mu = M/D_{\mathrm{arr}}, \qquad L = 2^a,$$

where $r \geq 2$ is an integer. The symbols used as vertex labels are strings in $[r]^a$, equivalently Q labels on each side.

Theorem 3.1 (Two-stage labeling)**.** *Suppose $M \leq s$, $\lambda = \lceil \log_2 s \rceil \geq 4$, all degrees are at most the positive integer Δ, and*

$$M \geq 4D_{\mathrm{arr}}\Delta\lambda^4. \tag{3.1}$$

There is a polynomial-time, polynomial-space deterministic labeling for which every pair-label class has load strictly below

$$B = \lceil (1+2/\lambda)\mu \rceil.$$

All decisions can be made using exact rational arithmetic.

Proof. **First stage: whole-row balancing.** Assign each row vertex, with weight its degree, to a currently least-loaded row label. Before an assignment the smallest load is at most the total assigned weight divided by Q, hence at most M/Q. The new item has weight at most Δ. Consequently every final row-label load R_j satisfies

$$R_j \le M/Q + \Delta. \tag{3.2}$$

Vertices are not split and no edge is changed.

Second stage: independent column labels. Temporarily assign each whole column vertex an independent uniform label in $[Q]$. Let w_{vj} be the number of edges from column vertex v to rows carrying label j. Then

$$0 \le w_{vj} \le \Delta, \qquad \sum_v w_{vj} = R_j.$$

For a fixed cell (j,k) its load is

$$X_{jk} = \sum_v w_{vj} \mathbf{1}_{\{\mathrm{label}(v)=k\}}, \qquad \mathbb{E}X_{jk} = R_j/Q \le \mu + \Delta/Q.$$

Independence is asserted for column vertices, not for individual parallel edges.

Set

$$\eta = \lambda^{-2}, \qquad z = 1 + \eta/\Delta.$$

For $0 \le w \le \Delta$,

$$z^w - 1 \le (w\eta/\Delta)e^{\eta} \le (w\eta/\Delta)(1+2\eta).$$

The last inequality follows, for $0 \le \eta \le 1/2$, from $e^{\eta} \le (1-\eta)^{-1} \le 1 + 2\eta$. The product formula for the moment generating function gives

$$\ln \mathbb{E}z^{X_{jk}} = \sum_v \ln\left(1 + \frac{z^{w_{vj}} - 1}{Q}\right) \tag{3.3}$$

$$\le \frac{\eta}{\Delta}(1+2\eta)(\mu + \Delta/Q). \tag{3.4}$$

Since $\mu \ge 4\Delta\lambda^4$ and $Q \ge 1$,

$$\mu + \Delta/Q \le \mu(1 + 1/(4\lambda^4)).$$

Also $\ln z \ge (\eta/\Delta)(1-\eta/2)$. Therefore

$$B\ln z - \ln \mathbb{E}z^{X_{jk}} \ge \frac{\mu\eta}{\Delta}\left[(1+2/\lambda)(1-\eta/2) - (1+1/(4\lambda^4))(1+2\eta)\right] \tag{3.5}$$

$$\ge \frac{\mu}{\Delta\lambda^3} \ge 4\lambda. \tag{3.6}$$

For completeness, the second inequality holds for every real $\lambda \ge 4$: after subtracting $1/\lambda$ from the bracket and multiplying by $4\lambda^6$, the result is

$$P(\lambda) = 4\lambda^5 - 10\lambda^4 - 4\lambda^3 - \lambda^2 - 2.$$

Writing $\lambda = x + 4$ gives the exact positive-coefficient certificate

$$P(x+4) = 1262 + 2360x + 1551x^2 + 476x^3 + 70x^4 + 4x^5.$$

Now consider

$$\Phi = \sum_{j,k} z^{X_{jk} - B}.$$

Condition (3.1) implies $D_{\mathrm{arr}} \le M \le s \le 2^{\lambda}$. Thus

$$\mathbb{E}\Phi \le D_{\mathrm{arr}} e^{-4\lambda} \le \exp((\ln 2 - 4)\lambda) < 1.$$

Exact derandomization. Suppose some column labels have already been chosen. Let A_{jk} be their contribution to cell (j,k). Apart from the common factor z^{-B}, the conditional potential is the rational number

$$\Psi = \sum_{j,k} z^{A_{jk}} \prod_{v \text{ unassigned}} \left(1 - \frac{1}{Q} + \frac{z^{w_{vj}}}{Q}\right). \tag{3.7}$$

Try every possible label of the next column vertex and choose one minimizing this expression. The average of the Q candidate values is its previous value, so the potential does not increase. At termination $\Phi < 1$. If some $X_{jk} \geq B$, its summand would be at least one. Hence every $X_{jk} < B$.

There are polynomially many cells and label choices because $D_{\text{arr}} \leq s$. All exponents are at most $M \leq s$; the rational base has polynomial bit length. Products over the explicitly described vertices therefore have polynomial bit length, and rational comparison is exact. Zero-degree vertices can be handled separately. This proves the claimed construction and resource bounds. □

3.1 Padding and the original orientation construction

Round the maximum pair-label load upward to a multiple H_{pad} of L. By integrality, $X_{jk} \leq B-1 < (1+2/\lambda)\mu$, so

$$H_{\text{pad}} \leq (1 + 2/\lambda)\mu + L.$$

There is no missing additive ceiling term in this inequality.

Corollary 3.2 (Packet and group counts)**.** *Under Theorem 3.1, also suppose* $2L \leq \Delta$*. Put*

$$\kappa = \frac{H_{\text{pad}} D_{\text{arr}}}{M}.$$

Then

$$\kappa \leq 1 + 2/\lambda + \frac{L D_{\text{arr}}}{M} \leq 1 + \frac{3}{\lambda}. \tag{3.8}$$

The source's orientation packet construction gives at most $\kappa M/L$ *packets and at most* $\kappa M/r$ *real ports on either side in each coordinate-symbol group. At a retained valuation node* $i < u$*, with* $D = L/2^i$*, there are at most* $2\kappa M/D$ *mode families, and a child has at most* $2\kappa M/r$ *real edges.*

Proof. Condition (3.1) and $L \leq \Delta/2$ give $LD_{\text{arr}}/M \leq 1/(8\lambda^4)$, proving (3.8). Pad each of the D_{arr} ordered label-pair classes to H_{pad} slots.

Fix the unordered pair of symbols in each coordinate. If a' coordinates have distinct symbols, there are $2^{a'}$ possible orientations. Each packet takes $2^{a-a'}$ slots from each corresponding ordered class. The a binary slot bits record the orientations in unequal coordinates and the copies in equal coordinates. Divisibility of H_{pad} by 2^a makes this assembly exact. The total number of packets is $H_{\text{pad}} D_{\text{arr}}/L = \kappa M/L$.

For any coordinate, a particular symbol appears in exactly a $1/r$ fraction of the padded ordered label classes on each side. Hence each real-port group has size at most $H_{\text{pad}} D_{\text{arr}}/r = \kappa M/r$. The source creates a child from the corresponding row and column groups, giving at most $2\kappa M/r$ edges.

At valuation i, each packet has $2^{i+1} = 2L/D$ column tiles. There is one mode family per relevant tile, so the total is bounded by $(\kappa M/L)(2L/D) = 2\kappa M/D$. □

The old sufficient condition was $M \geq 2D_{\text{arr}}^2\Delta$, obtained by bounding the sum of squared cell deviations. The new theorem improves its dependence on D_{arr}, in addition to removing slack from the padding count.

Proposition 3.3 (Why the new power of D_{arr} is natural)**.** *For arbitrary bipartite multigraphs, a uniform conclusion* $\max X_{jk} \leq (1 + o(1))M/D_{\text{arr}}$ *requires* $M/D_{\text{arr}} \geq (1 - o(1))\Delta$ *in general.*

Proof. Take a component consisting of two vertices joined by Δ parallel edges, and add disjoint edges as needed. That entire bundle belongs to one ordered pair-label cell under any whole-vertex labeling. Its load is at least Δ. □

This is a limitation of near-balanced whole-vertex packing. It is not a SAT lower bound, nor a claim that the polylogarithmic factor in (3.1) is optimal.

4 Precision: the reused exact taper and the new guard

Use the positive linear taper

$$q_i = h(u-i) \quad (0 \le i < u), \qquad q_i = 0 \quad (i \ge u).$$

There are $L/2^i$ grid points of valuation i. Consequently the exact number of Hermite conditions is

$$hW(a,u), \qquad W(a,u) = \sum_{i=0}^{u-1} \frac{L}{2^i}(u-i) = L(2u-2+2^{1-u}).$$

The smallest sufficient integer slope is the earlier refinement

$$h = \left\lceil \frac{M(2L-1)+1}{W(a,u)} \right\rceil. \tag{4.1}$$

It satisfies $h \le 3M/u$ when $M \ge u$, and $h = (1+o(1))M/u$ uniformly when $u \to \infty$, $u/M \to 0$, and $L \to \infty$.

For a node t with $v_2(t) = i < u$, the source's exact Hermite loss, as used in [5], is

$$B_i = D\sum_{j=1}^{a-i} \frac{j}{2^j}(q_i - q_{i+j}), \qquad D = 2^{a-i}.$$

For a linear taper,

$$B_i = hD\sum_{j=1}^{a-i} \frac{j\min(j,u-i)}{2^j} \le hD\sum_{j\ge 1} \frac{j^2}{2^j} = 6hD.$$

The multiplier of the z^j coefficient in recovery has valuation $-B_i - j(a+1-i)$. Thus the exact sufficient precision is $p + B_i + (q_i-1)(a+1-i)$; a convenient larger integral prescription is

$$P_t = p + 6hD + hu(a+1). \tag{4.2}$$

This preserves the earlier leading constant six and strictly increases precision along recursive paths.

The source and earlier schedules used $a_d = 2\sum_{j\le d} u_j$. Define instead

$$g = \lceil 8\log_2(\lambda+1) \rceil, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)g, \qquad a_{-1} = 0. \tag{4.3}$$

For every retained $i < u_d$,

$$D = 2^{a_d - i} \ge 2^{a_{d-1}+g+1}. \tag{4.4}$$

Lemma 4.1 (Inherited precision is negligible)**.** *Suppose active calls have $M > s/\lambda$, depth $O(\log\lambda)$, $1 \le u \le \lambda$, $a \le \lambda$, and uniformly $u \to \infty$. With (4.1), (4.2), and (4.3),*

$$P_t \le (6+o(1))MD/u$$

uniformly over active calls and retained nodes.

Proof. Since $h \le 3M/u$, one precision increment is at most $18s2^a + 3s(a+1)$. Summing preceding increments, using $P_0 \le s$ and nondecreasing a, gives the convenient coarse bound

$$p \le 22s\lambda^2\, 2^{a_{d-1}}$$

for sufficiently large λ. By (4.4),

$$\frac{p}{MD/u} \le 11\lambda^4\, 2^{-g} = O(\lambda^{-4}),$$

and

$$\frac{hu(a+1)}{MD/u} \le O(\lambda^2 2^{-g}) = O(\lambda^{-6}).$$

Finally $6hD = (6+o(1))MD/u$. All estimates are uniform under the stated hypotheses. □

Only a logarithmic guard is needed. This is a distinct change from counting the Hermite conditions more accurately.

5 An integer capacity schedule

Fix

$$r = 54, \qquad b = 9, \qquad H = \lceil s/\lambda \rceil, \qquad \Delta = \left\lceil s^{9/10} \right\rceil.$$

Calls with $M \le H$ are leaves. Define

$$d_{\max} = \lceil \log_2 \lambda \rceil, \qquad G = (d_{\max} + 1)g, \tag{5.1}$$

$$A = \max\{A \in \mathbb{Z}_{\ge 0} : 4r^{2A}\Delta\lambda^6 \le s\}, \qquad K = A - G. \tag{5.2}$$

If the set in (5.2) is empty, $\lambda < 4$, or $K \le 0$, use direct contraction. Otherwise set

$$u_d = \left\lfloor \frac{8K}{9^{d+1}} \right\rfloor, \qquad a_d = \sum_{j=0}^{d} u_j + (d+1)g. \tag{5.3}$$

If an actual active call has $u_d = 0$, use direct contraction at that call. This is a finite-size guard, not an additional global requirement on $u_{d_{\max}}$.

Lemma 5.1 (Packing and depth without circular assumptions)**.** *Every non-fallback active call has $a_d \le A$, satisfies the packing hypotheses, and has children with at most $(1 + 3/\lambda)M/27$ edges. No active call requires a depth larger than $d_{\max}$.*

Proof. The geometric sum gives $\sum_{j\ge 0} u_j \le K$. At any depth $d \le d_{\max}$,

$$a_d \le K + (d_{\max} + 1)g = A.$$

The cap then gives

$$4r^{2a_d}\Delta\lambda^4 \le s/\lambda^2 < M.$$

Also $r^{2a_d} \ge 2^{2a_d} = L^2$, so $4L^2\Delta\lambda^6 \le s$. Since $\Delta^3\lambda^6 \ge s$, it follows that $2L \le \Delta$. Theorem 3.1 and Corollary 3.2 apply.

For $\lambda \ge 4$, $(1 + 3/\lambda)/27 < 1/2$. Inductively a depth-d node has at most $s2^{-d}$ edges, or an ancestor has already terminated. A child beyond $d_{\max}$ therefore has fewer than s/λ edges and is a leaf. This closes the induction using only depths where $a_d \le A$ was established. □

Lemma 5.2 (Uniform active taper growth)**.** *For sufficiently large s, no active call takes a parameter fallback, and uniformly over active depths*

$$u_d = (1 + o(1))\varepsilon_*\lambda\, 9^{-d}, \qquad \varepsilon_* = \frac{2}{45\log_2 54}. \tag{5.4}$$

In particular the smallest active u_d tends to infinity.

Proof. Maximality of A, together with $\log_2 \Delta = (9/10)\log_2 s + o(1)$, gives

$$A = \frac{\lambda}{20\log_2 r} - \frac{3}{\log_2 r}\log_2 \lambda + O(1).$$

Also $G = O((\log\lambda)^2)$, so

$$K = \frac{\lambda}{20\log_2 r} - O((\log\lambda)^2).$$

For $\lambda \ge 78$, the child ratio is at most $1/26$. Any active depth consequently satisfies $d < \log_{26}\lambda$. Its unfloored taper is at least a positive constant times

$$\lambda\, 9^{-d} \ge \lambda^{\,1-\ln 9/\ln 26}, \qquad 1 - \frac{\ln 9}{\ln 26} = 0.3256109658\ldots > 0.$$

Thus the absolute floor error below one is a uniform relative $o(1)$. Substituting the leading term of K proves (5.4). The cap exists and $K > 0$ eventually, and the actual active tapers are then positive. □

Remark 5.3 (An important finite-size distinction). It would be incorrect to require $u_{d_{\max}} \ge 1$ before accepting the schedule: $d_{\max}$ is deliberately loose, and that artificial taper can vanish even as s tends to infinity. Only actual active depths, which are bounded by $\log_{26}\lambda$, need positive taper.

5.1 All parameter decisions are exact

For integers $s \geq 2$, compute

$$\lambda = (s-1).\,\mathrm{bit_length}, \qquad d_{\max} = (\lambda - 1).\,\mathrm{bit_length},$$

and

$$g = ((\lambda+1)^8 - 1).\,\mathrm{bit_length}\,.$$

The integer Δ is the least integer with $\Delta^{10} \geq s^9$. Integer comparisons determine A from (5.2). No floating-point logarithm is used to choose a feasible schedule.

For readers preferring fixed parameters, the choices

$$(r, b, \varepsilon) = (64, 10, 1/135) \quad \text{or} \quad (54, 9, 2/259)$$

with $u_d = \lfloor \varepsilon\lambda/b^d \rfloor$ and the same additive guard give the coefficients listed in Section 1. For the second choice, strict packing slack follows from the exact integer inequality $54^{45} < 2^{259}$. The adaptive cap avoids leaving a fixed fraction of the available packing exponent unused.

6 Reusing the factorial Cauchy modes

We restate the needed content of [4] to make compatibility checkable. For a retained node, write $T = t + 2^{a+1}\zeta$, $v_2(t) = i$. In one column tile,

$$y = y_0 + 2^{i+1}j, \qquad 0 \leq j < J, \qquad J = D/2.$$

For a crossing row, put

$$d_0 = t - y_0 + x, \qquad z_x = \frac{2^{i+1}}{d_0 + 2^{a+1}\zeta}, \qquad c_x = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)(d_0 + 2^{a+1}\zeta)}.$$

The source provides $d_0 \neq 0$, $v_2(d_0) \leq i$, and $z_x, c_x \in 2\mathcal{O}$. The entry is $c_x/(1 - z_x j)$.

Define

$$A_b(z) = \prod_{h<b} \frac{z - 2h}{1 - hz}, \qquad B_b(j) = \prod_{h<b} \frac{j - h}{1 - 2hj}.$$

The elementary identity

$$(1 - bz)(1 - 2bj) - (z - 2b)(j - b) = (1 - 2b^2)(1 - zj)$$

telescopes to

$$\frac{1}{1 - zj} = \sum_{b=0}^{J-1} \frac{(1 - 2b^2)A_b(z)B_b(j)}{(1 - bz)(1 - 2bj)} \qquad (0 \leq j < J).$$

The remainder is zero because $B_J(j) = 0$. Move the integer factorial from the column falling product to the row coefficient:

$$\ell_b(j) = \frac{(1 - 2b^2)\binom{j}{b}}{\prod_{h=0}^{b}(1 - 2hj)}, \tag{6.1}$$

$$r_b(x) = c_x\, b!\, \frac{A_b(z_x)}{1 - bz_x}. \tag{6.2}$$

Then

$$W_{xy}(T) = \sum_{b=0}^{J-1} r_b(x)\ell_b(j).$$

The source feature is integral and all displayed denominators are units; no factorial is inverted in the residue ring.

To justify the second factorial valuation over the extension, set $w = z_x/2$ and $w_0 = 2^i/d_0 \in \mathbb{Z}_{(2)}$. Direct subtraction gives

$$w - w_0 \in 2^{a+1-i}\mathcal{O}.$$

For $b < 2^{a+1-i}$, the product $\prod_{h<b}(w-h)$ has valuation at least $v_2(b!)$: at each binary precision, count the integers h in the appropriate residue class of w_0. This is valid for every relevant $b < J$. Consequently

$$v_2(r_b(x)) \geq w_b := 1 + b + 2v_2(b!) = 3b + 1 - 2s_2(b). \tag{6.3}$$

The near-prime-subring congruence is essential; the assertion is not made for arbitrary extension elements w.

The new packing retains the same tile, slot, and source/target identities. The adaptive schedule changes neither $J = D/2$ nor the congruence exponent $a + 1 - i$. Thus (6.3) remains available with no extra families. For a mode tuple β, terms with $\sum_{f,b} w_b\beta_{f,b} \geq P_t$ vanish modulo 2^{P_t}. The remaining tuples factor into the original child contractions.

7 Entropy and the complete time bound

Let $N(P, c)$ count nonnegative mode-exponent tuples of total weight below P with at most c families. Extending finite tiles to all $b \geq 0$ only increases this count. For every $\theta > 0$,

$$\ln N(P,c) \leq \theta P + c\sum_{b\geq 0} -\ln(1 - e^{-\theta w_b}).$$

Because $w_b = 3b + O(\log(b+2))$,

$$\sum_{b\geq 0} -\ln(1 - e^{-\theta w_b}) = \left(\frac{\pi^2}{18} + o(1)\right)\frac{1}{\theta} \quad (\theta \downarrow 0).$$

This follows by a Riemann-sum comparison and $\int_0^\infty -\ln(1-e^{-x})\,dx = \pi^2/6$; the finitely many smallest weights and their logarithmic perturbations are lower order. Optimizing θ gives

$$\ln N(P,c) \leq \left(\frac{\pi\sqrt{2}}{3} + o(1)\right)\sqrt{cP}. \tag{7.1}$$

At our calls,

$$c \leq (2 + o(1))M/D, \qquad P_t \leq (6 + o(1))MD/u.$$

Ceilings in the family envelope are negligible because $M/D \to \infty$ uniformly. Optimizing the stated upper envelopes for c and P_t, the entropy saddle satisfies $\theta = O(\sqrt{u}/D) = O(\lambda^{-15/2})$, using $D \geq 2^{g+1}$ and $u \leq \lambda$. Hence the $o(1)$ in (7.1) is uniform. We obtain

$$\ln N_{\text{call}} \leq (2\pi\sqrt{2/3} + o(1))\, M/\sqrt{u}. \tag{7.2}$$

Theorem 7.1 (Improved normalized contraction certificate)**.** *Under the normalized interface of Section 2, the two-stage packing, additive guard, and adaptive capacity schedule, combined with the inherited contraction identities and factorial modes, compute the exact contraction deterministically in polynomial space and time*

$$(s+\mathcal{L})^{O(1)}2^{(C_*+o(1))s/\sqrt{\log_2 s}}, \qquad \boxed{C_* = \frac{9\pi}{4\ln 2}\sqrt{15\log_2 54}}. \tag{7.3}$$

Proof. At depth d, the product of the small padding overheads satisfies

$$(1 + 3/\lambda)^d = 1 + o(1)$$

uniformly for $d = O(\log \lambda)$. Thus

$$M_d \le (1 + o(1)) s\, 27^{-d}.$$

Lemma 5.2 gives $u_d = (1 + o(1))\varepsilon_* \lambda\, 9^{-d}$. Summing the logarithms of the branching bounds along a path using (7.2) yields

$$\left(\frac{2\pi\sqrt{2/3}}{\sqrt{\varepsilon_*}} \sum_{d \ge 0} (3/27)^d + o(1) \right) \frac{s}{\sqrt{\lambda}}.$$

The geometric sum is $9/8$. Dividing by $\ln 2$ and substituting $\varepsilon_* = 2/(45 \log_2 54)$ gives precisely (7.3).

The remaining operations do not change this coefficient. The number of interpolation points, nodes, and child choices is polynomial in s. Polynomial overhead across $O(\log \lambda)$ levels contributes $O(\lambda \log \lambda)$ to the logarithm. Leaves have at most $H = \lceil s/\lambda \rceil$ edges, giving a lower-order $O(s/\lambda)$ exponent.

For the arithmetic and entry routines, $a \le A = O(\lambda)$ and the precision bound of Lemma 4.1 is polynomial in s. In fact $L \le s^{1/(20 \log_2 54) + o(1)}$, so $M(2L - 1) < s^4$ eventually. The original residue-field extension has enough distinct interpolation points. Existing vertices retain their degrees and new packet vertices have at most $2L \le \Delta$ ports. The original packet-entry cost $\exp(O(L \log s))$ and original entry allowance $\exp(O(s^{9/10}))$ both have lower-order logarithms. The earlier streamed finite-difference routine can also be used, but is not necessary for this coefficient.

Each new packet entry is computed directly from its own creation-time packet data, mode tuple, interpolation value, precision, and local kernel. Previously existing tensors remain separate vertices and are evaluated from their own original or creation-time descriptions. Thus entry routines do not compose recursively. New entry evaluation has the stated $\exp(O(L \log s))$ bound, while retained descriptions are immutable and shared by reference. The number of recursive alternatives depends polynomially on s and not on $\mathcal{L}$; the description-dependent work per call retains a fixed polynomial factor in $s + \mathcal{L}$.

The depth-first tuple enumeration, exact rational packing calculations, coordinate arithmetic, and recursive child accumulation use polynomial space. Each tensor keeps its creation-time canonical representatives. The small-size and zero-taper fallbacks occur only for a finite range of s, by the uniform growth proof, and do not affect the asymptotic bound. □

7.1 A certified numerical enclosure

The supplied checker uses rational alternating-series bounds for Machin's identity and positive-series bounds for logarithms. It certifies

$$94.748202259534 < C_* < 94.748202259535.$$

Floating-point arithmetic is used only for convenient table displays.

The adaptive schedule attains this leading coefficient with one algorithm. It is stronger than merely showing that a family of algorithms approaches the fixed-parameter infimum as a parameter tends to a boundary. No global optimality assertion is made.

8 Algorithm specification and integration

The specification below identifies every replacement while preserving the original exact-algebra interfaces.

1. Normalize the root to the declared $s, \mathcal{L}, P_0, \ell, \Delta$ interface. Construct the original unramified coefficient extension and retain the creation-precision convention.
2. Compute $\lambda, g, d_{\max}, G, A, K$ using integer arithmetic. Use direct contraction when the root fallback applies.
3. At a call with M edges, return a direct contraction if $M \le H$. Otherwise compute u_d, a_d and use the finite fallback if $u_d = 0$.
4. Greedily balance whole row vertices among r^{a_d} labels. Assign whole column vertices by minimizing the exact conditional potential (3.7).

5. Pad the pair-label loads to a common multiple of L, then use the original orientation-and-copy assembly. Tag new tensor descriptions with their creation precision.
6. Use (4.1) for the Hermite taper. At each retained node use the sufficient precision (4.2).
7. Substitute the factorial Cauchy features and enumerate tuples of total certified weight below P_t. For a fixed tuple, form the unchanged source/target child contractions and evaluate them recursively.
8. Recover the Taylor data and then $F(0)$ by the original exact interpolation and Hermite formulas. Reduce to the precision of the call.

The natural integration point is the collision-packing subsection of `quantitative-contraction.tex`, followed by its parameter, precision, and running-time proofs. The source deformation, coefficient extension, and tensor-creation semantics remain part of the proof dependencies. The accompanying code implements the new labeling, packet assembly on finite examples, integer schedules, and local checks; it does not implement all eight steps as a production contractor.

9 Consequences for SAT and limits of the improvement

Applying the new contractor to the existing reductions preserves exact counting, determinism, polynomial space, and the general and fixed-density orders (1.1)–(1.2). The density-aware checks from [7, 8] can be retained, giving the same density-sensitive exponent envelope

$$(n+m)^{O_k(1)}2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}$$

with the improved contraction component. The scalar lift of [6] is compatible with the scalar special case $\ell = 1$.

A numerical coefficient in terms of n requires tracking the reduction's wire budget, heavy-variable branches, array enumeration, and all other leading terms. This report does not replace an unknown O_k-constant in a SAT statement by C_*. Nor does it compare C_* numerically with the different method in [10], whose coefficient is not specified.

For a concrete occurrence-based application, one may take the original bipartite variable–clause incidence graph. Handle $N = 0$ directly. If there are $N \geq 1$ literal occurrences, branch on variables occurring more than $\lceil (N+1)^{9/10} \rceil$ times. There are at most $N/(N+1)^{9/10}$ such variables. Within each branch retain every unfixed original incidence, including incidences in clauses whose restricted tensor is constant; this preserves the counting of all unfixed occurring variables. The remaining variable equality tensors and clause tensors satisfy the normalized degree cap with $s = N+1$ and can be evaluated in polynomial time. The branching cost is lower order, and the count precision is at most $N+1$ after unused variables have been removed. Thus for fixed width one gets the explicit occurrence-based certificate

$$\mathrm{poly}(n,m)\,2^{(C_*+o(1))(N+1)/\sqrt{\log_2(N+1)}}.$$

Unused declared variables contribute their known power of two afterward. This application illustrates the normalization; the new mathematical contribution is the contractor, not the elementary incidence encoding.

9.1 Why the logarithmic order has not changed

The changes above increase how much taper can be used and reduce the number of mode families. Their leading scales are still

$$c = \Theta(M/D), \qquad P = \Theta(MD/u), \qquad u = \Theta(\log s)$$

near the root. Therefore $\sqrt{cP} = \Theta(M/\sqrt{u})$ persists.

There is also a direct obstruction to improving this order solely by enumerating the same unrestricted mode tuples more selectively by degree. Suppose $c = \Theta(M/D)$ colors have modes up to $t = \lfloor \xi D/\sqrt{u} \rfloor$, with a

sufficiently small fixed $\xi > 0$. Since $w_b \le 3b + 1$, every zero-one choice among these ct coordinates has total weight at most $O(ct^2) \le P$ and total degree $ct \le M$. There are

$$2^{ct} = \exp(\Omega(M/\sqrt{u}))$$

such tuples. This is a count in the abstract enumeration model; it does not assert that all their coefficients are nonzero, or lower-bound SAT. It explains why merely adding a total-degree cap does not supply the missing logarithmic improvement.

10 Verification and artifact contents

The new packing proof, precision schedule, and adaptive-cap argument were checked through separate mathematical reviews during this work. Those reviews are analytic audits, not proof-assistant verification. Finite checks use exact integer or rational arithmetic and explicit exceptions rather than assertions, so optimized Python does not disable the checks.

Executed finite check	Count
Packing graph/parameter cases	240
Conditional formula compared with exhaustive label assignments	720
Complete label assignments inspected	4,256
Exact one-step averaging identities	480
Monotonic conditional-potential checks	480
Packet embeddings with padded marginal checks	48
Real-edge orientation identities	192
Invalid packing-input controls	8
Rational Chernoff-bracket instances	997
Exact Hermite taper instances	225
Ordinary rational finite Cauchy identities	360
Ordinary rational source/target valuation checks	2,040
Exact integer capacity-schedule instances	26
Cumulative integer taper-budget checks	1,080
Weighted mode-count instances	12

The positive-coefficient polynomial in (3.6) is an infinite-domain algebraic certificate, not merely a grid test. The numerical enclosure of C_* has rigorous rational remainder bounds. The separate rational Cauchy tests do not replace the extension-ring congruence proof in Section 6.

The small packing examples directly verify their initial-potential certificates. They are not claimed to satisfy the very large asymptotic sufficient size inequality. Likewise, moderate exact schedule examples can legitimately take the direct fallback. The eventual validity and uniform growth of active tapers are proved in Section 5; no practical crossover size is inferred.

Artifact	Purpose
`report.tex`, `report.pdf`	Complete theorem, proofs, comparison, and limitations
`code/packing.py`	Exact two-stage labeling and orientation-preserving packet assembly
`code/schedule.py`	Integer capacity schedule, taper, mode counts, and constant enclosure
`checks/check_packing.py`	Exhaustive and exact finite packing checks
`checks/check_analysis.py`	Analytic algebra certificates and local identities
`results/*.json`	Recorded verification outputs and comparison numbers

Artifact	Purpose
`provenance/*.json`	Source pins, archive hashes, changed-file records, and review scope
`README.txt`, `Makefile`	Reproduction and integration instructions
`MANIFEST.sha256.json`	Checksums of the released contents

Reproduction requires Python 3.10 or later for the checks and a standard LaTeX distribution for the PDF. The code has no third-party Python dependencies. Run the two check scripts from the archive root; build the PDF with the provided Makefile. No downloaded research script is required to execute the new checks.

11 Further research questions

Sum surviving modes without visiting them individually. A smaller logarithmic order requires more than the improvements proved here. Can the actual structured mode sum be evaluated in polynomial space with logarithm $o(M/\sqrt{u})$? A useful theorem must account for the large inherited precision and for high degrees in any compressed mode network. An exact representation alone does not establish that such a network can be contracted efficiently.

Use the degree structure of SAT reductions. The normalized certificate allows degrees as large as $s^{9/10}$. Many reduction networks have considerably smaller effective degrees. A degree-sensitive packing and entry-cost analysis may improve explicit coefficients further. Its benefit should be stated with the actual input and child degree bounds, rather than silently substituting a smaller exponent into the generic theorem.

Reduce the reduction's incidence-encoding entropy. For arbitrary clause density, the existing subset encodings introduce $O(n \log\log n)$ wires. Can a different exact representation reduce this to $O(n)$ while preserving local entry evaluation, bounded degrees, and the creation-precision semantics? Density-sensitive checks already help in particular regimes, but do not by themselves remove the unrestricted $\log\log n$ factor.

Sharpen the whole-vertex packing theorem. The parallel-edge example shows why a factor $D_{\mathrm{arr}}\Delta$ is natural. Can the λ^4 sufficient margin be reduced while preserving simple exact derandomization and controlled description sizes? This is unlikely to change the leading coefficient of the adaptive schedule, which already pays only lower-order budget for polylogarithmic margins, but could reduce its exceptionally large finite threshold.

Formalize the actual replacement interfaces. A formal verification should connect whole-vertex conditional expectations, orientation packing, the new schedule's depth induction, and the factorial mode features to the original machine-level definitions. The scope should include immutable creation-time tensor lifts. Verifying only a numerical inequality for C_* would leave the algorithmic content unverified.

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